TENNESSEE TECH UNIVERSITY · DEPARTMENT OF COMPUTER SCIENCE

Exam Timetabling at Scale

Benchmarking thirteen algorithms on the ITC 2007 examination track

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Assign every exam to a (period, room) pair — *minimizing a single penalty term, subject to hard constraints.*

OBJECTIVE · fitness = hard × 100 000 + soft

HARD

× 100 000

No two exams of one student share a period. Room capacity respected. Period duration budget respected. Period- and room-specific hard rules honoured.

SOFT (PENALISED)

× 1

Exams in-a-row / same-day / n-apart. Mixed-duration exams sharing a room. Large exams late in the session. Period and room weighting.

NP-HARD (GRAPH COLORING FORMULATION) · NO POLYNOMIAL-TIME ALGORITHM KNOWN · FEASIBILITY FIRST, THEN MINIMIZE SOFT PENALTY

Four research questions

AGENDA

One benchmark, thirteen algorithms — four directions this deck defends.

Q1

SCALING

How does each algorithm's runtime scale from 50 to 1 000 exams?

Q2

PARETO

Where does each sit on the quality-vs-runtime frontier?

Q3

SENSITIVITY

Which tunable knob matters most, per algorithm?

Q4

RELIABILITY

Does CP-SAT degrade gradually or fall off a cliff?

A portfolio of thirteen

APPROACH

One C++20 solver — four algorithm families, thirteen members.

Construction

1 algo

Greedy (DSatur)
Saturation-degree graph coloring — a single pass.

Trajectory

7 algos

Tabu Search
Feasibility-first swap + room-only moves

Simulated Annealing
Multi-neighbourhood geometric cooling

Kempe Chain
Conflict-chain period swaps, SA acceptance

Great Deluge
Linearly decaying acceptance level

LAHC
Late-acceptance hill climbing

ALNS
Adaptive destroy-and-repair

GVNS
Variable neighbourhood + SA kernel

Population

4 algos

Genetic Alg.
Memetic GA — Kempe mutation + DSatur crossover

ABC
Artificial Bee Colony — multi-move bees

WOA
Whale Optimization — spiral + encircling

HHO+
Harris Hawks + Lévy flight + local refinement

Exact / Hybrid

1 algo

CP-SAT
OR-Tools BnB — 300 s cap in IP-only batch, 120 s in the main cross-algorithm batch.

Fast enough to benchmark

METHODOLOGY

Engineering that makes the comparison fair and the numbers reproducible.

01 Delta evaluation

Every local-search move computes Δcost in O(k), not O(n²).

Single biggest speed-up. Every trajectory method leans on it; without Δ, 10 000-iter runs simply would not complete.

02 Warm-start chains

--init-solution pipes one algorithm's output into the next.

Chain-finder searches over chains: successive halving ladder + prefix cache + 1-point crossover selects alns → kempe → tabu.

03 Auto-tuner

4-phase pipeline — screen, tune, chain-hunt, multi-seed validate.

Global mode normalises per-dataset and uses geometric mean. Gated updates: no single dataset may regress more than 15 %.

BENCHMARK

ITC 2007 Examination Track — 8 real instances (242 to 1 096 exams) + synthetic generator. Every algorithm runs on every set; 3 seeds each; final table reports mean ± std.

How each algorithm scales

METHODOLOGY · RUNTIME COMPLEXITY

Theoretical Θ per iteration · empirical mean runtime at n = 1 000.

THEORETICAL Θ PER ITERATION

Algorithm	Inner loop	Θ (iter)
Greedy (DSatur)	single pass	Θ(n + m)
Tabu	swap + Δcost	Θ(k)
SA	sample + Δcost	Θ(k)
Kempe	chain + swap	Θ(n + c)
Great Deluge	sample + Δcost	Θ(k)
LAHC	Δcost + history	Θ(k)
ALNS	destroy + repair	Θ(k log n)
GVNS	shake + SA kernel	$\Theta(N \cdot T_1 \cdot k)$
GA (memetic)	mutate + eval	$\Theta(n^2)$
ABC	P bees × move	Θ(P·k)
WOA	pop × n	Θ(P·n)
HHO+	Lévy + refine	Θ(P·n)
CP-SAT	branch & bound	exp. worst

EMPIRICAL FLOOR · MEAN RUNTIME @ n = 1 000

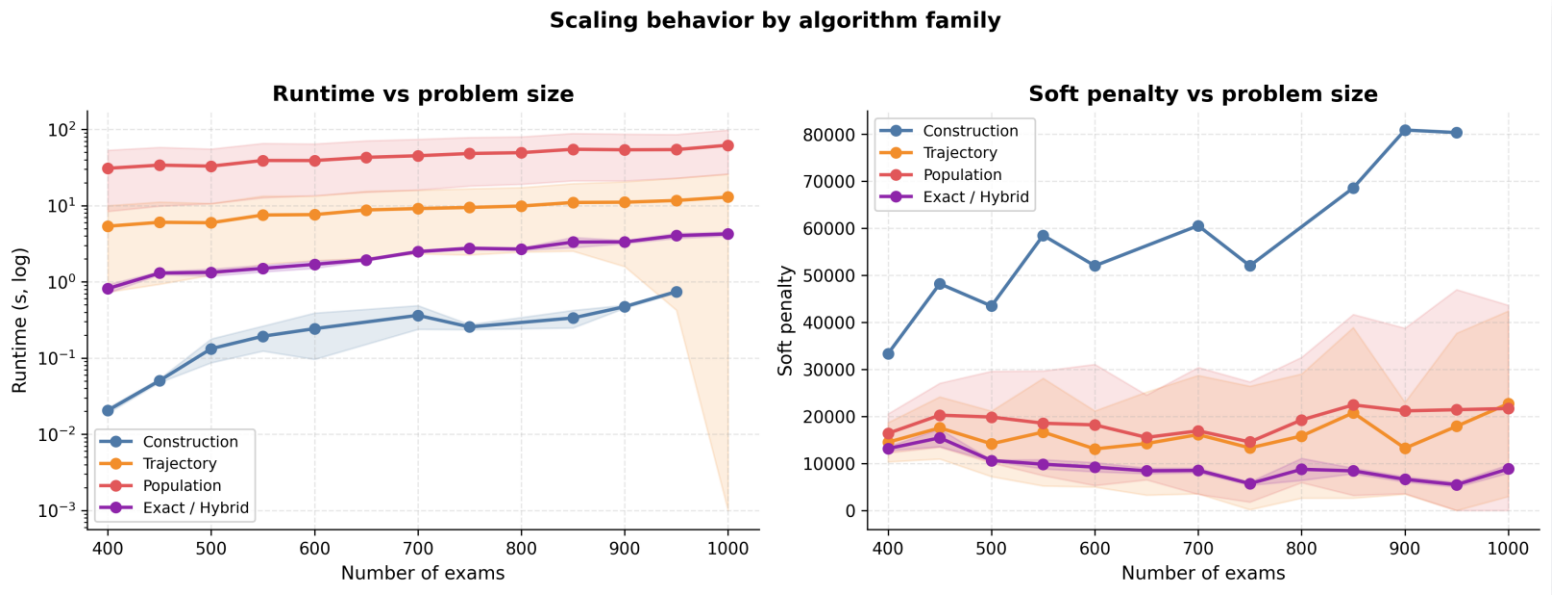
Algorithm	Mean (s)	Feasible?
Greedy (DSatur)	0.76	no
Tabu	16.75	yes
SA	2.39	yes
Kempe	15.60	yes
Great Deluge	2.08	yes
LAHC	2.48	yes
ALNS	40.36	yes
GVNS	10.98	yes
GA (memetic)	5.11	yes
ABC	86.10	yes
WOA	92.68	yes
HHO+	64.36	yes
CP-SAT	4.26	yes

Empirical from colab_batch_scaling, 3 seeds at n = 1 000. Θ treats T, P, k, c as independent parameters.

Runtime scales cleanly — until it doesn't

RQ1 · SCALING

Most algorithms are linear or near-linear in n; Construction stays cheap but its quality does not.



Runtime (log-log, left) and soft penalty (right) vs problem size, grouped by algorithm family.

TRAJECTORY

Nearly linear — Δ-evaluation keeps per-move cost flat.

CONSTRUCTION

Cheapest by 2 orders of magnitude; quality blows up on dense sets.

POPULATION

Dominated by pop × evaluator — steepest runtime slope.

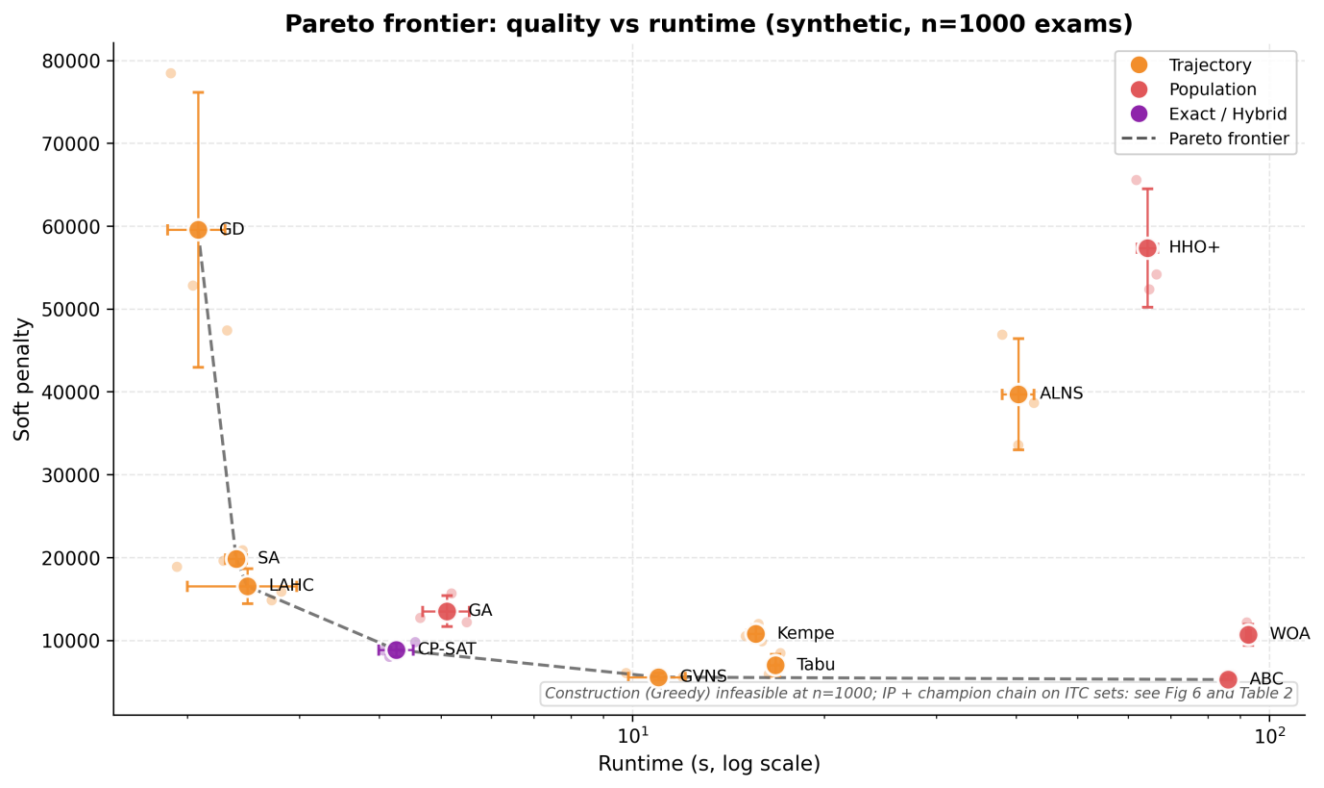
EXACT / HYBRID

Omitted — returns incumbents (no proof) above n ≈ 900; not a scaling comparison.

The Pareto frontier

RQ2 · QUALITY VS RUNTIME

Trajectory methods own the bottom-left corner — best quality per second spent.



Pareto frontier, synthetic n = 1000. Bottom-left dominates.

FRONTIER MEMBERS

Tabu

Best quality / time on small–medium

GVNS

Close second; lowest family rank

Kempe

Tight runner-up on dense sets

ABC

Only Population member near frontier

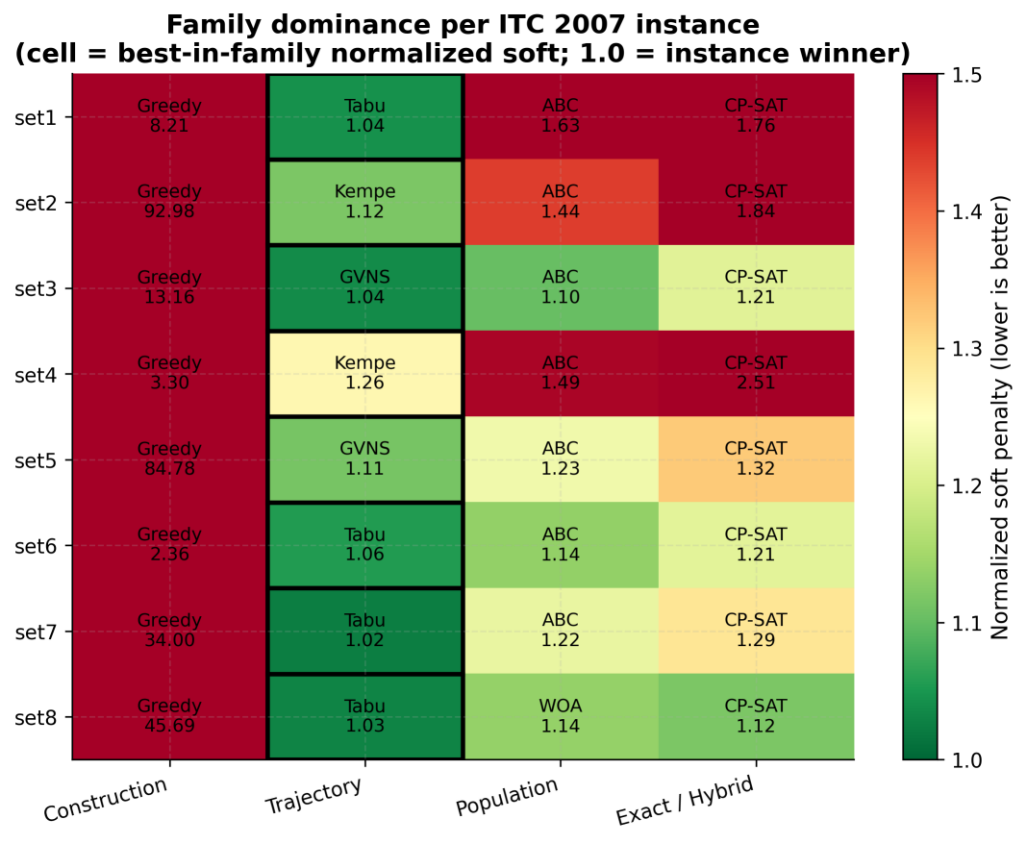
Greedy

Speed extremum — quality extremum too

Who wins where

RESULTS · FAMILY DOMINANCE

Trajectory takes every instance. Tabu leads the family; ABC dominates Population 7-of-8.



Normalised soft penalty; green = in-family winner per instance. F-Rank (right) = mean rank within family (lower = better); Population has 4 members, Trajectory 7.

WITHIN-FAMILY LEADERS

1 TRAJECTORY
Tabu
F-Rank 2.12 · Wins 4

2 POPULATION
ABC
F-Rank 1.12 · Wins 7

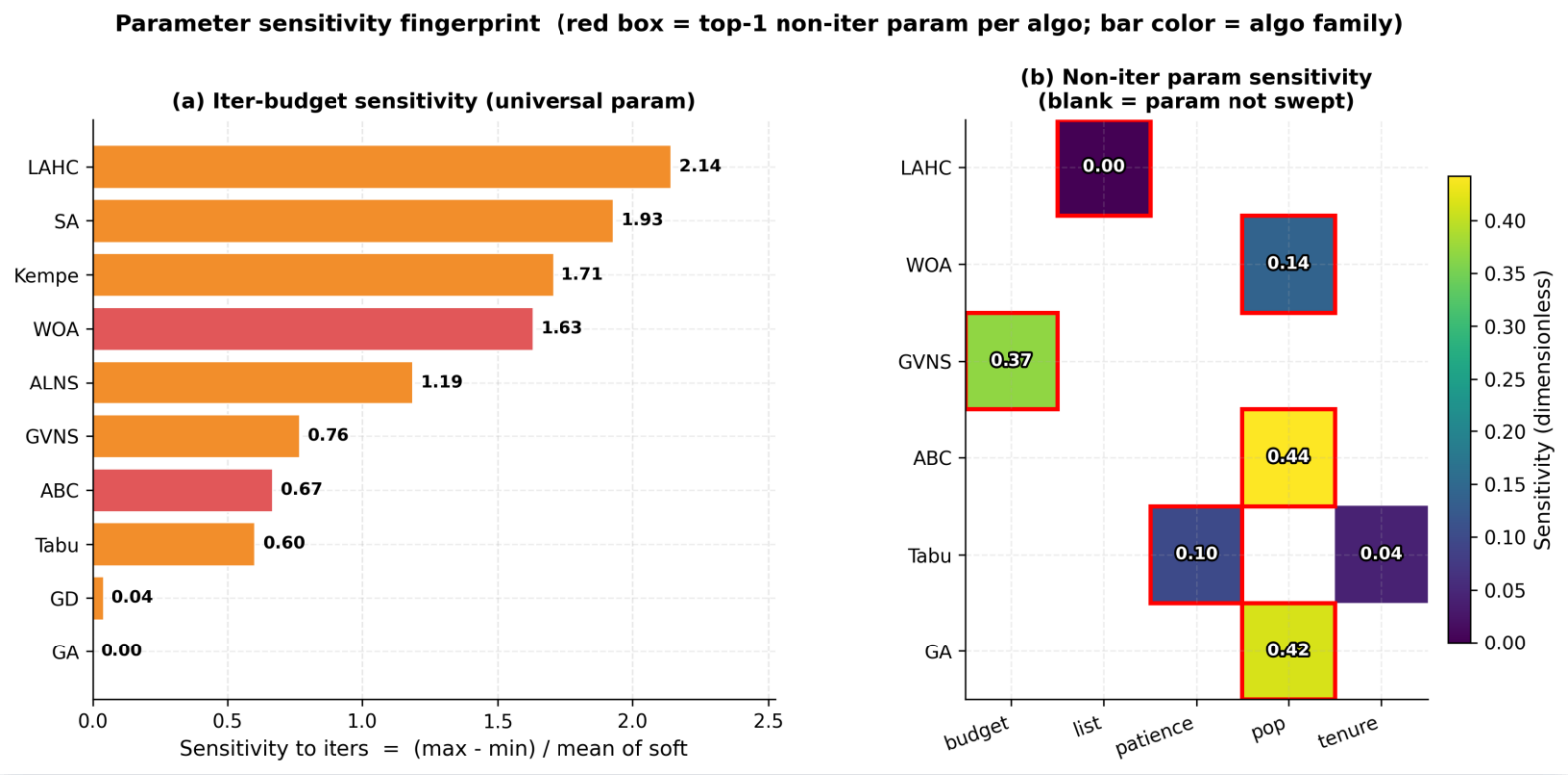
— CONSTRUCTION
Greedy
F-Rank — · Wins —

— EXACT / HYBRID
CP-SAT
F-Rank — · Wins —

Which knob matters?

RQ3 · PARAMETER SENSITIVITY

Sensitivity = (max – min) / mean soft penalty across a 1-D sweep.



Left panel: iters sensitivity (universal). Right panel: top non-iters knob per algorithm.

DOMINANT KNOB

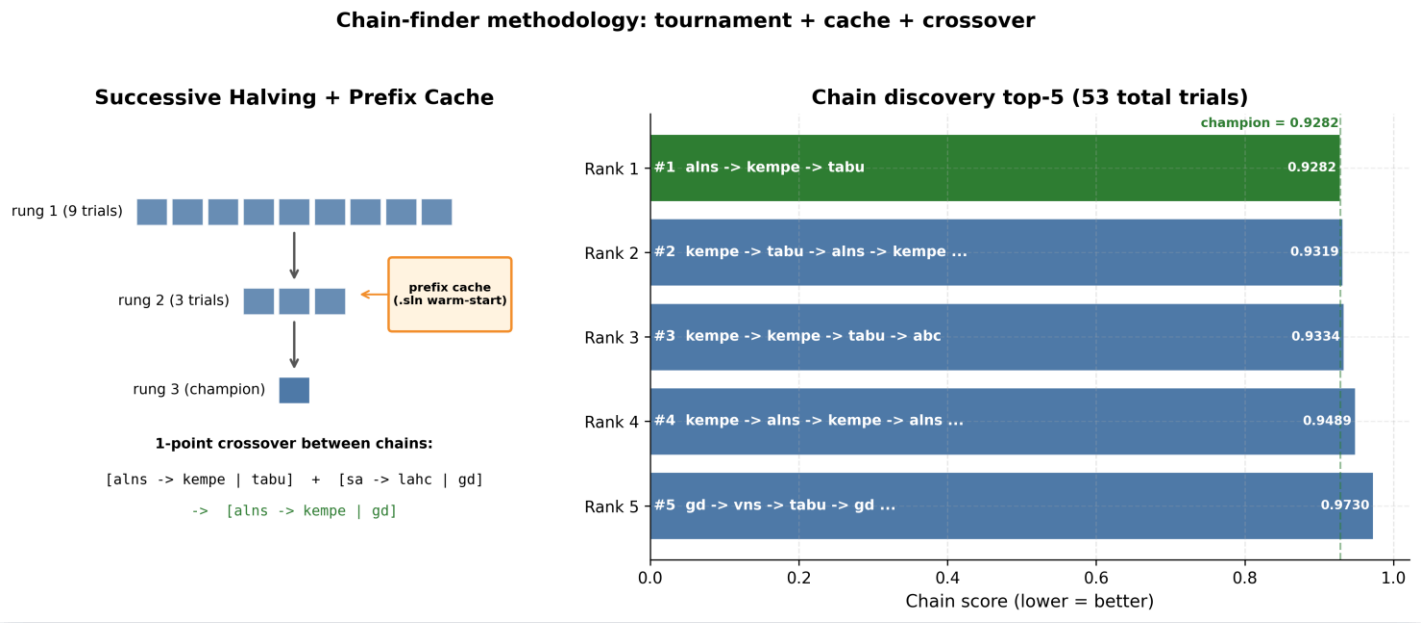
ALNS	swing %
SA	cooling
GA	pop size
Tabu	tenure
Kempe	periods
ABC	bees

Iters matter everywhere, but one per-family knob dominates the remaining variance — tune it first.

The champion chain

NOVEL CONTRIBUTION · CHAIN-FINDER

alns → kempe → tabu — discovered by tournament, validated across all 8 sets.



Successive-halving ladder + prefix cache + 1-point crossover → top-5 discovered chains.

TOP-5 CHAINS (SCORE = NORMALISED GEOMEAN)

#1	alns → kempe → tabu	0.928
#2	kempe → tabu → alns → …	0.932
#3	kempe → kempe → tabu → abc	0.933
#4	kempe → alns → kempe → …	0.949
#5	gd → vns → tabu → gd → alns	0.973

Chain-finder selects the shortest chain within 0.5 % of the best score — trades 3.5× fewer total iterations for a ≈ 0.5 % hit. ALNS diversifies, Kempe repairs period structure, Tabu polishes.

A reliability cliff, not a slope.

CP-SAT solves every instance ≤ 870 exams and fails every instance ≥ 934.

SOLVED

242–870

EXAMS

All four instances solved to a non-improving incumbent inside the 300 s IP-batch budget — no timeouts in this band.

900

EXAM THRESHOLD

TIMED OUT

934–1 096

EXAMS

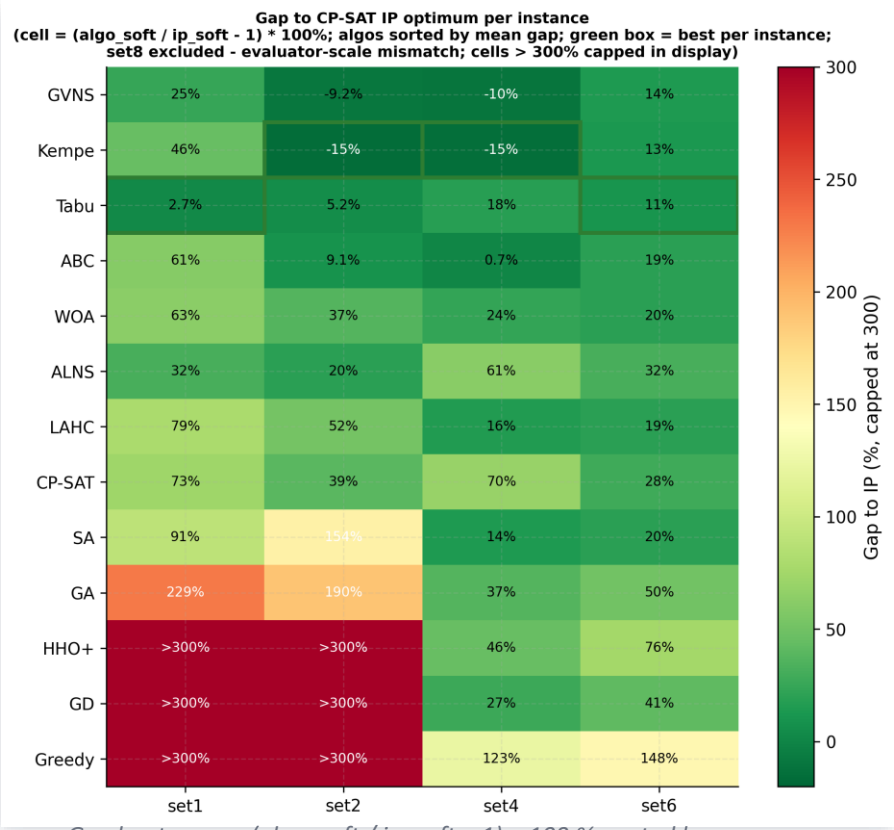
No proof returned inside the 300 s IP-batch budget for any instance in this band.

Above the cliff, heuristics are not merely faster — they are the only ones that finish.

Gap to proven optimum

RQ4 · CLOSING THE GAP

On 7 instances with a CP-SAT proof, top trajectory methods stay within single-digit percent (set8 excluded).



Gap heatmap — (algo_soft / ip_soft − 1) × 100 %, sorted by mean gap.

HEADLINE GAPS

Tabu	**+2.7 %**
GVNS	**+4.1 %**
Kempe	**+8.5 %**
ABC	**+9.3 %**
ALNS	**+12.1 %**
SA	**+45.7 %**

Tabu — the within-family winner on the heatmap — is also closest to the proven optimum. On instances where CP-SAT can return a proof, trajectory methods come within a few percent for a tiny fraction of the runtime.

Full leaderboard

RESULTS · ALL 13 ON 8 INSTANCES

Mean soft penalty over 3 seeds, across all eight ITC 2007 sets, sorted by mean rank.

Algo	s1	s2	s3	s4	s5	s6	s7	s8	Rank
GVNS	10 614	2 364	21 941	25 305	5 424	31 634	9 357	13 970	**1.88**
Tabu	8 740	2 740	25 558	33 163	5 789	30 733	8 043	12 972	**2.38**
Kempe	12 383	2 211	25 189	23 999	10 975	31 449	10 938	14 923	**3.50**
ABC	13 663	2 841	23 287	28 365	5 995	33 081	9 592	14 339	3.75
WOA	13 862	3 578	30 832	34 807	11 086	33 160	10 386	14 332	5.88
CP-SAT	14 715	3 628	25 611	47 735	6 428	35 336	10 137	14 084	6.25
LAHC	15 267	3 952	33 516	32 594	12 893	33 057	12 979	15 202	6.62
SA	16 249	6 613	40 625	32 225	29 308	33 144	21 832	19 954	8.00
ALNS	11 252	3 117	159 k	45 419	43 657	36 662	16 186	17 420	8.25
GA	27 975	7 544	43 496	38 595	13 046	41 681	13 158	22 823	9.38
GD	41 973	130 k	173 k	35 701	330 k	39 173	115 k	526 k	11.00
HHO+	54 546	123 k	154 k	41 154	264 k	48 776	152 k	323 k	11.12
Greedy	68 678	183 k	277 k	62 720	412 k	68 625	267 k	575 k	13.00
IP†	8 509	2 604	—	28 162	—	27 710	—	25‡	—

Trajectory · Population · Construction · Exact / Hybrid

† IP† = CP-SAT full-converge (reference row, not a 14th algo). ‡ set8 = evaluator-scale anomaly.

What this work contributes

TAKEAWAYS

01

Trajectory methods dominate.

Tabu, GVNS, and Kempe own the Pareto frontier. Tabu wins 4-of-8 intra-family slots and sits within +2.7 % of the CP-SAT optimum.

02

Population methods are a distant second.

ABC is the best of the Population family (7-of-8 intra-family wins) but lags Trajectory by 5–15 % on tight instances.

03

Exact solvers have a reliability cliff at ~900 exams.

CP-SAT solves every instance under 870 exams and times out on every instance over 934. Heuristics are not merely faster — they are the only option on the upper half of ITC 2007.

04

Warm-start chains beat monoculture.

Chain-finder discovers alns → kempe → tabu as the champion: diversify, then repair, then polish — shorter and better than any single algorithm alone.

05

A single C++20 solver with Δ-eval makes this comparison honest.

Every algorithm runs under identical timing, neighborhood, and evaluator code — no cross-implementation bias.

Thank you.

Questions · Discussion · Code review

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github.com/Dialovos/exam-scheduling-algos-analyze